

# Week 6, Differential Calculus & Coordinate Geometry

→ Partial Derivatives

→ Chain Rule of Partial Derivative

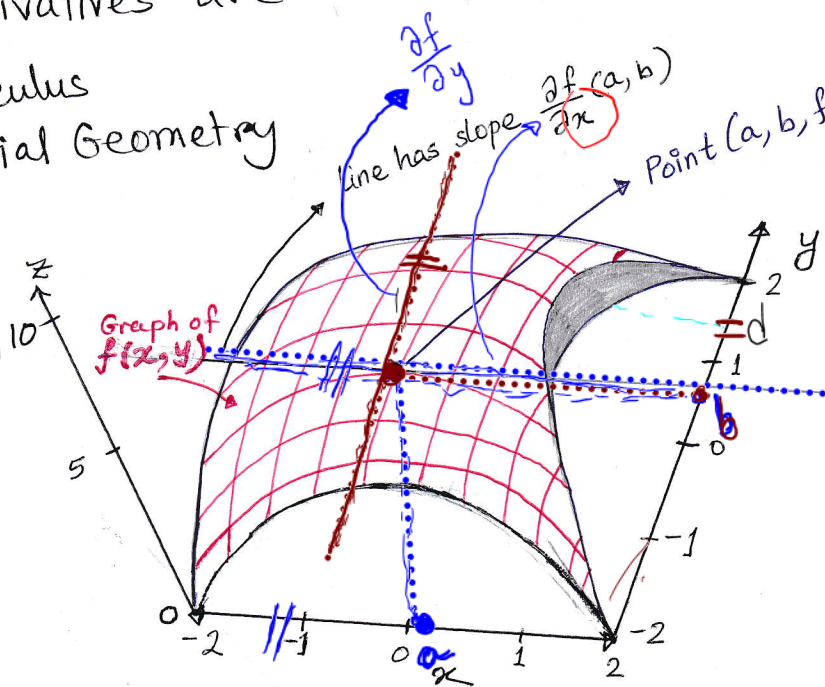
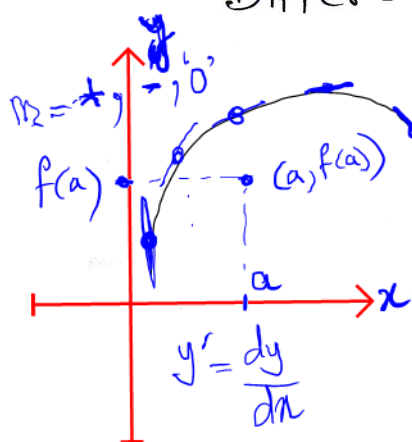
## PARTIAL DERIVATIVES

In mathematics a partial derivative of a function of several variables is its derivative w.r.to (with respect to) one of those variables, with the others held constant (as opposed to the total derivative, in which all variables are allowed to vary)

Partial Derivatives are used in:

→ Vector Calculus

→ Differential Geometry



$$\begin{aligned} x &= a \\ y &= b \\ (x, y) &= (a, b) \end{aligned}$$

$$\begin{aligned} z &= f(x, y) \\ &= f(a, b) \end{aligned}$$

$$\begin{aligned} (x, y, z) &= (a, b, f(a, b)) \end{aligned}$$

$x$ , variable  
 $y$  → constant

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$y$  - variable  
 $x$  - constant

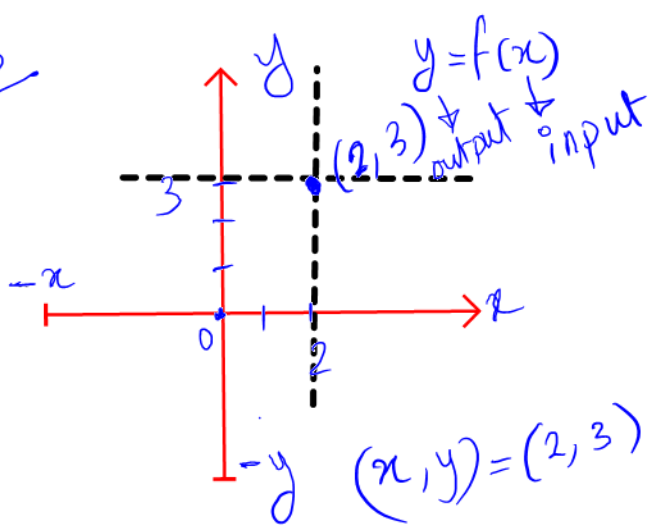
$$\frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Compare the limit def<sup>n</sup> of  
one variables  
 $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$   
or  $\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{(x - a)}$$

$$\begin{aligned} \Delta x &= x_2 - x_1 \\ &= x - a \end{aligned}$$

2D



$$y = f(x)$$

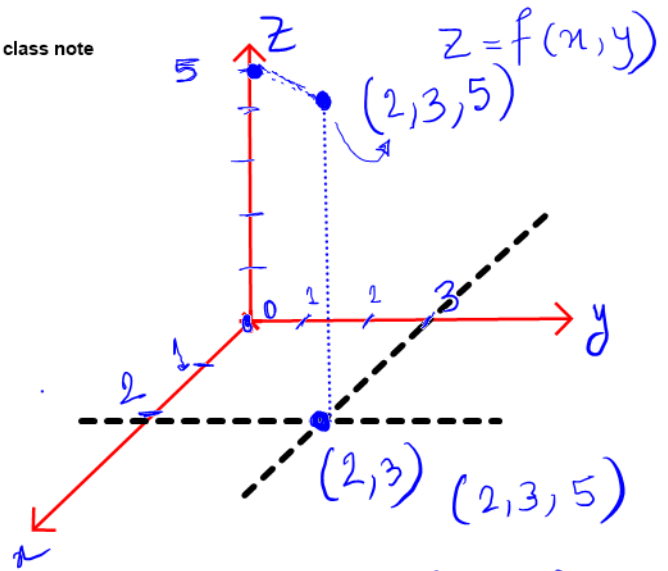
$$y' = f'(x)$$

$$y'' = f''(x)$$

$$y''' = f'''(x)$$

$$\vdots \quad \frac{dy}{dx}$$

Consultation class note



$$z = f(x, y)$$

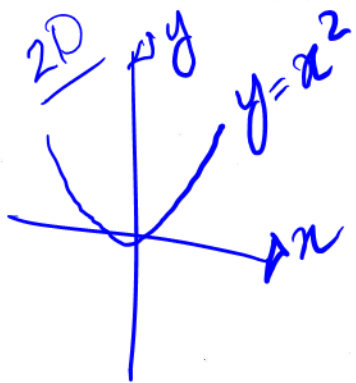
output input

$$z = f(x, y) \quad \text{variable} = ??$$

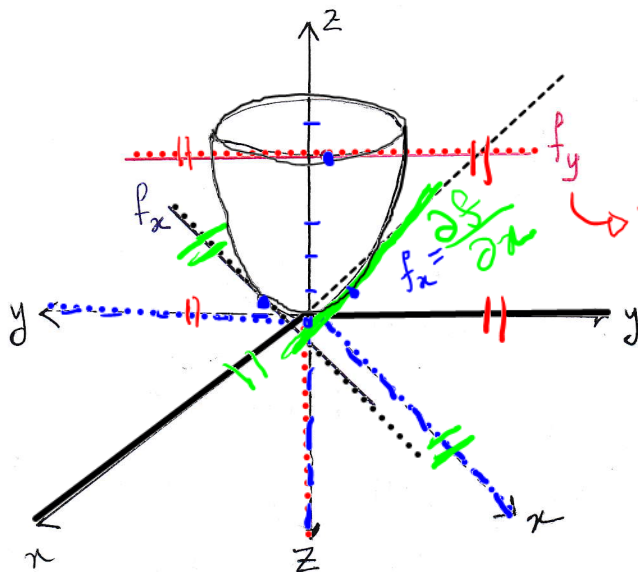
$$\begin{matrix} z' & \rightarrow & f_x, f_y \\ z'' & \rightarrow & f_{xx}, f_{yy} \end{matrix}$$

$$\left. \begin{matrix} f_x = \frac{\partial f(x, y)}{\partial x} \\ f_y = \frac{\partial f(x, y)}{\partial y} \end{matrix} \right\} \begin{matrix} f_{xy} \\ f_{yx} \end{matrix}$$

$\frac{\partial f}{\partial x}$   
 $\frac{\partial f}{\partial y}$

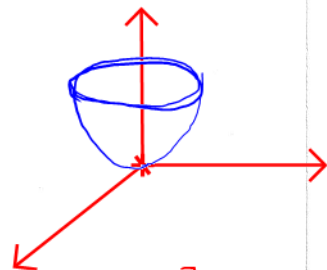


3D



paraboloid:  
 $z = x^2 + y^2$   
 $f(x, y) = x^2 + y^2$

$\frac{\partial f}{\partial y}$



$\frac{df(x)}{dx} \rightarrow 2D$

3D Mathematical Notation:

$\frac{\partial}{\partial x} f(x, y) = f_x(x, y) = f_x \rightarrow \text{slope}$

$\frac{\partial}{\partial y} f(x, y) = f_y(x, y) = f_y \rightarrow \text{slope}$

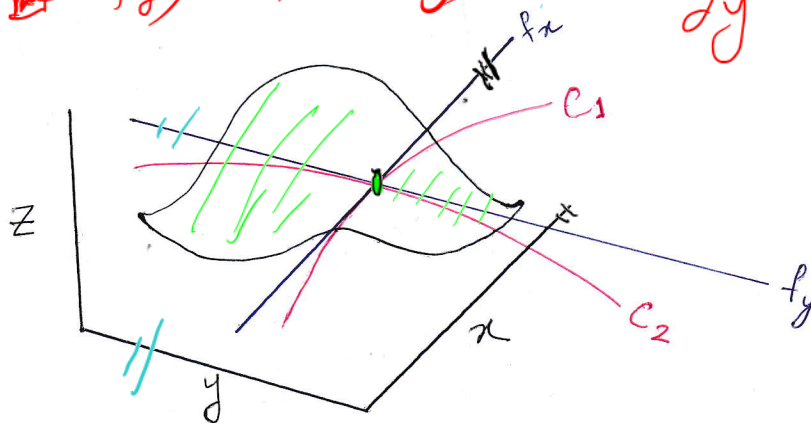
$f'(x)$   
 $f''(x)$   
 $f_{xx}, f_{yy},$   
 $f_{xy}$

$\frac{dy}{dx} \rightarrow 2D$

Geometric Significance

$\frac{\partial f}{\partial x} = f_x(x, y) = f_x(x, y)$

$\frac{\partial f}{\partial y} = f_y(x, y)$



### Exercise

1] Let  $f(x, y) = 3x^3y^2$ . Find the followings

$$a) f_x(x, y) = \frac{\partial}{\partial x} (3x^3y^2) = 3(3x^2)y^2 = 9x^2y^2$$

$$b) f_y(x, y) = \frac{\partial}{\partial y} (3x^3y^2) = 3x^3(2y) = 6x^3y$$

$$c) f_x(x, 1) = 9x^2(1^2) = 9x^2 \quad \text{--- refer (a)}$$

$$d) f_y(1, 2) = 6(1)^3(2) = 12 \quad \text{--- refer (b)}$$

2] Let  $f(x, y) = 4x^2 - 2y + 7x^4y^5$ . Find the following:

a)  $f_{xx}$

$$f(x, y) = 4x^2 - 2y + 7x^4y^5$$

$$f_x(x, y) = 8x - 0 + 28x^3y^5$$

$$f_{xx}(x, y) = 8 + 84x^2y^5$$

b)  $f_{yy}$

$$f(x, y) = 4x^2 - 2y + 7x^4y^5$$

$$f_y(x, y) = 35x^4y^4 - 2$$

$$f_{yy}(x, y) = 35x^4 \cdot 4y^3 - 0 \\ = 140x^4y^3$$

c)  $f_{xy}$

$$f(x, y) = 4x^2 - 2y + 7x^4y^5$$

$$f_x(x, y) = 8x + 28x^3y^5$$

$$f_{xy}(x, y) = 0 + 28x^3 \cdot 5y^4 \\ = 140x^3y^4$$

d)  $f_{yx}$

$$f(x, y) = 4x^2 - 2y + 7x^4y^5$$

$$f_y(x, y) = 35x^4y^4 - 2$$

$$f_{yx}(x, y) = 35y^4 \cdot 4x^3 \\ = 140x^3y^4$$

$(x, y)$   
2!

$x, y$   
 $y, x$

$$f_{xy} = f_{yx}$$

3

2] Let  $f(x, y) = 4x^2 - 2y + 7x^4y^5$ . Find the following:

a)  $f_{xx}$

$$\begin{aligned} f(x, y) &= 4x^2 - 2y + 7x^4y^5 \\ f_x(x, y) &= 8x - 0 + 28x^3y^5 \\ f_{xx}(x, y) &= 8 + 84x^2y^5 \end{aligned}$$

b)  $f_{yy}$

$$\begin{aligned} f(x, y) &= 4x^2 - 2y + 7x^4y^5 \\ f_y(x, y) &= -2 + 35x^4y^4 \\ f_{yy}(x, y) &= 140x^4y^3 \end{aligned}$$

c)  $f_{xy}$

$$\begin{aligned} f(x, y) &= 4x^2 - 2y + 7x^4y^5 \\ f_x(x, y) &= 8x + 28x^3y^5 \\ f_{xy}(x, y) &= 0 + 28x^3 \cdot 5y^4 \\ &= 140x^3y^4 \end{aligned}$$

d)  $f_{yx}$

$$\begin{aligned} f(x, y) &= 4x^2 - 2y + 7x^4y^5 \\ f_y(x, y) &= -2 + 35x^4y^4 \\ f_{yx}(x, y) &= 35y^4 \cdot 4x^3 \\ &= 140x^3y^4 \end{aligned}$$

$$z = f(x, y)$$

1<sup>st</sup> der:  $f_x$   $f_y$

2<sup>nd</sup> der:  $f_{xx}$   $f_{yy}$   
 $f_{xy} = f_{yx}$

$$\left. \begin{aligned} \frac{d}{dx} C &= 0 \\ \frac{d}{dx} 2x &= 2 \end{aligned} \right\}$$

$$f(x, y, z) = x^3y^5z^7 + xy^2 + y^3z$$

$f_{zyy}$

$$\begin{aligned} f_z &= x^3y^5 \cdot 7z^6 + 0 + y^3 \\ &= 7x^3y^5z^6 + y^3 \end{aligned}$$

$$f_{zy} = 35x^3y^4z^6 + 3y^2$$

$$\begin{aligned} f_{zyy} &= 35x^3 \cdot 4y^3z^6 + 6y \\ &= 140x^3y^3z^6 + 6y \end{aligned}$$

$f_{yzy}$

$$\begin{aligned} f_y &= x^3z^7 \cdot 5y^4 + x^2y + 3y^2z \\ &= 5x^3y^4z^7 + x^2y + 3y^2z \end{aligned}$$

$$f_{yz} = 5x^3y^4 \cdot 7z^6 + 0 + 3y^2 = 35x^3y^4z^6 + 3y^2$$

$$f_{yzy} = 35x^3z^6 \cdot 4y^3 + 6y = 140x^3y^3z^6 + 6y$$

$$y = f(x) \quad 2D$$

$$z = f(x, y) \quad 3D$$

$$w = f(x, y, z) \quad 4D$$



$$w = f(x, y, z)$$

4D  
[3] Let  $f(x, y, z) = x^3 y^5 z^7 + xy^2 + y^3 z$ , Find the followings:

a)  $f_{xy}$

$$f = x^3 y^5 z^7 + xy^2 + y^3 z$$

$$f_x = 3x^2 y^5 z^7 + y^2$$

$$f_{xy} = 15x^2 y^4 z^7 + 2y$$

b)  $f_{yz}$

$$f = x^3 y^5 z^7 + xy^2 + y^3 z$$

$$f_y = 5x^3 y^4 z^7 + 2xy + 3y^2 z$$

$$f_{yz} = 35x^3 y^4 z^6 + 3y^2$$

c)  $f_{xz}$

$$f = x^3 y^5 z^7 + xy^2 + y^3 z$$

$$f_x = 3x^2 y^5 z^7 + y^2$$

$$f_{xz} = 21x^2 y^5 z^6$$

d)  $f_{zz}$

$$f = x^3 y^5 z^7 + xy^2 + y^3 z$$

$$f_z = 7x^3 y^5 z^6 + y^3$$

$$f_{zz} = 42x^3 y^5 z^5$$

e)  $f_{zyy}$

$$f = x^3 y^5 z^7 + xy^2 + y^3 z$$

$$f_z = 7x^3 y^5 z^6 + y^3$$

$$f_{zy} = 35x^3 y^4 z^6 + 3y^2$$

$$f_{zyy} = 140x^3 y^3 z^6 + 6y$$

$f_{zxy}$

$$f = x^3 y^5 z^7 + xy^2 + y^3 z$$

$$f_z = 7x^3 y^5 z^6 + y^3$$

$$f_{zx} = 21x^2 y^5 z^6$$

$$f_{zxy} = 105x^2 y^4 z^6$$

$$f_{zxy} = f_{zyx}$$

h)  $f_{xxyz}$

$$f = x^3 y^5 z^7 + xy^2 + y^3 z$$

$$f_x = 3x^2 y^5 z^7 + y^2$$

$$f_{xx} = 6xy^5 z^7$$

$$f_{xxy} = 30xy^4 z^7$$

$$f_{xxyz} = 210xy^4 z^6$$

$$= f_{xyx}$$

$$= f_{xzy}$$

$$= f_{yxz}$$

$$= f_{yzx}$$

[4] Let  $f(x, y, z) = \sqrt{xy} + \ln(x^2 z^3) - x \tan(z)$

Compute  $f_x, f_{xy}, f_{xyz}, f_z$

(15n) a)  $f = \sqrt{xy} + \ln(x^2 z^3) - x \tan(z)$

$f_x = \frac{1}{2\sqrt{xy}}(y) + \frac{1}{x^2 z^3}(2x z^3) - \tan z$

$f_x = \frac{1}{2} x^{-1/2} y^{1/2} + \frac{2}{x} - \tan z$

b)  $f_{xy} = \frac{1}{2} x^{-1/2} \frac{1}{2} y^{1/2-1} + 0 - 0$   
 $= \frac{1}{4} x^{-1/2} y^{-1/2}$

$f_{xy} = \frac{1}{4\sqrt{xy}}$

c)  $f_{xyz} = 0$

d)  $f = \sqrt{xy} + \ln(x^2 z^3) - x \tan(z)$

$f_z = 0 + \frac{1}{x^2 z^3} x^2 3z^2 - x \sec^2 z$

$= \frac{3}{z} - x \sec^2 z$

$y = f(x) \rightarrow 2D$

$z = f(x, y) \rightarrow 3D$

$w = f(x, y, z) \rightarrow 4D$

$u = f(x_1, x_2, \dots, x_n, x_{n+1})$   
 $\rightarrow n+1D$

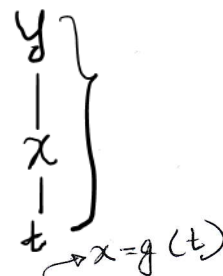
## CHAIN RULE of PARTIAL DERIVATIVES

Recall: if  $y = f(x)$ ,  $x = g(t)$

then  $\frac{dy}{dt} = [f(g(t))]'$   
 $= \frac{dy}{dx} \cdot \frac{dx}{dt}$

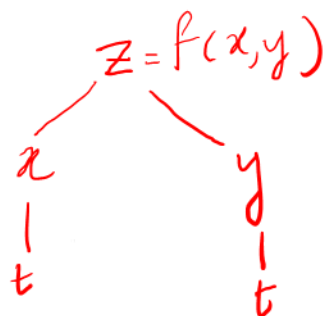
$y = f(x) \rightarrow \frac{df(x)}{dx}$

$\frac{dg(t)}{dt}$



The Chain Rule (Case 1) Suppose  $z = f(x, y)$  is a differentiable function of  $x$  and  $y$ , where  $x = g(t)$  and  $y = h(t)$  are both differentiable functions of  $t$ . Then  $z$  is a differentiable function of  $t$  and

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$



OR  $\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$

$z = f(x, y)$

Examples The pressure  $p$  (in kilopascals), volume  $V$  (in liters) and temperature  $T$  (in kelvins) of a mole of an ideal gas are related by the equation  $PV = 8.31T$ . Find the rate at which the pressure is changing when the temperature is  $300\text{ K}$  and increasing at a rate of  $0.1\text{ K/s}$  and the volume is  $100\text{ L}$  and increasing at a rate of  $0.2\text{ L/s}$ .

$\frac{dV}{dt}$



If  $t$  = time elapsed in seconds

$$T = 300$$

$$V = 100$$

$$\frac{dT}{dt} = 0.1$$

$$\frac{dV}{dt} = 0.2$$

Given:

$$PV = 8.31 T$$

$$P = 8.31 \frac{T}{V}$$

$$\frac{\partial P}{\partial T} = \frac{8.31}{V}$$

$$\frac{\partial P}{\partial V} = 8.31 T (-1) V^{-2} = - \frac{8.31 T}{V^2}$$



$$\begin{aligned} \frac{dP}{dt} &= \frac{\partial P}{\partial T} \cdot \frac{dT}{dt} + \frac{\partial P}{\partial V} \cdot \frac{dV}{dt} \\ &= \frac{8.31}{V} \cdot \frac{dT}{dt} - \frac{8.31 T}{V^2} \cdot \frac{dV}{dt} \end{aligned}$$

$$= \frac{8.31}{100} \cdot (0.1) - \frac{8.31(300)}{100^2} (0.2)$$

$$= -0.04155$$

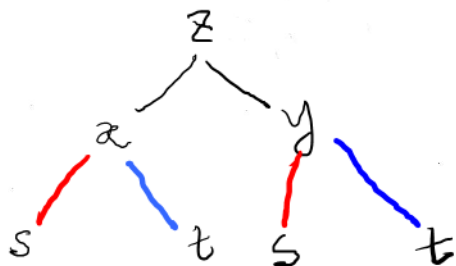
The pressure is decreasing at a rate of about  $0.042 \text{ kPa/s}$

**The chain Rule (Case 2)** Suppose that  $z = f(x, y)$  is a differentiable function of  $x$  and  $y$ , where  $x = g(s, t)$  and  $y = h(s, t)$  are differentiable functions of  $s$  and  $t$ .

Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$



Example If  $z = e^x \sin y$ , where  $x = st^2$  and  $y = s^2 t$ , find  $\frac{\partial z}{\partial s}$  and  $\frac{\partial z}{\partial t}$ .

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

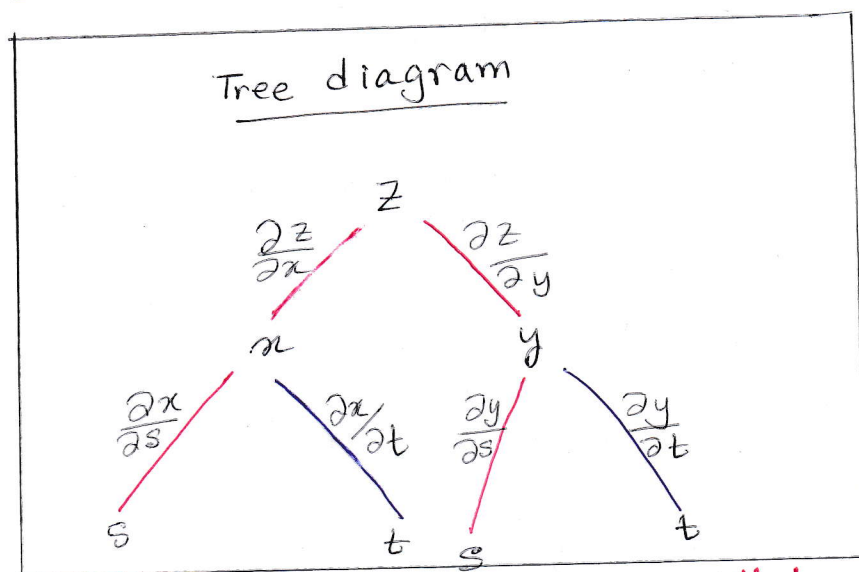
$$= (e^x \sin y) (t^2) + (e^x \cos y) (2st)$$

$$= t^2 e^{st^2} \sin(s^2 t) + 2st e^{st^2} \cos(s^2 t)$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$= (e^x \sin y) (2st) + (e^x \cos y) (s^2)$$

$$= 2st e^{st^2} \sin(s^2 t) + s^2 e^{st^2} \cos(s^2 t) = s e^{st^2} (2t \sin(s^2 t) + s \cos(s^2 t))$$



Chain Rule (General formula): Suppose that  $u$  is differentiable function on the  $n$  variables  $x_1, x_2, \dots, x_n$  and each  $x_j$  is a differentiable function of the  $m$  variables  $t_1, t_2, \dots, t_m$ . Then  $u$  is a function of  $t_1, t_2, \dots, t_m$  and

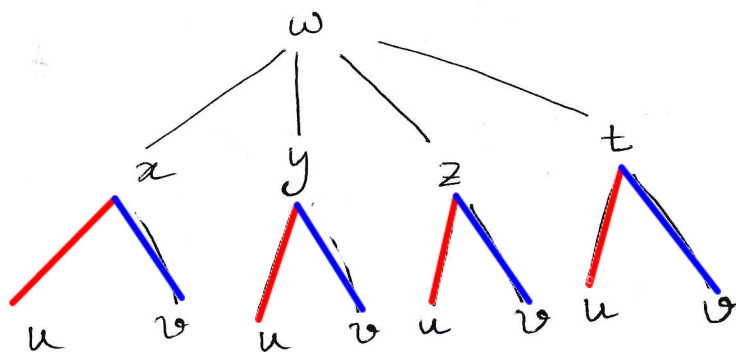
$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}$$

Example Write out the chain rule for the case where  $w = f(x, y, z, t)$  and  $x = x(u, v)$ ,  $y = y(u, v)$ ,  $z = z(u, v)$  and  $t = t(u, v)$ .

$$n=1, \quad m=2$$

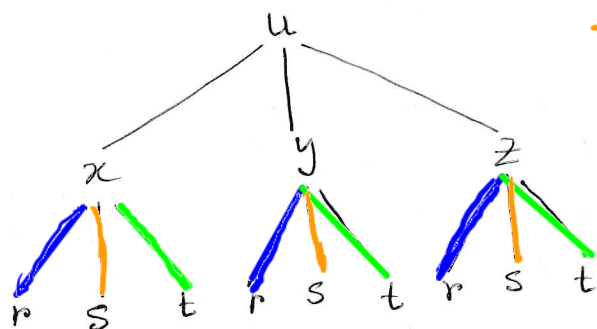
$$\rightarrow \frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial u} + \frac{\partial w}{\partial t} \frac{\partial t}{\partial u}$$

$$\rightarrow \frac{\partial w}{\partial v} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial v} + \frac{\partial w}{\partial t} \frac{\partial t}{\partial v}$$



Example If  $u = x^4 y + y^2 z^3$ , where  $x = r s e^t$ ,  $y = r s^2 e^{-t}$  and  $z = r^2 s \sin t$ , find the value of  $\frac{\partial u}{\partial s}$  when  $r=2, s=1, t=0$ .

Tree diagram **Given,  $x=2, y=2, z=0$ .**



With the help of tree diagram:

$$\rightarrow \frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial s}$$

$$= (4x^3 y)(r e^t) + (x^4 + 2y z^3)(2r s e^{-t}) + (3y^2 z^2)(r^2 \sin t)$$

$$= (64)(2) + (16)(4) + (0)(0)$$

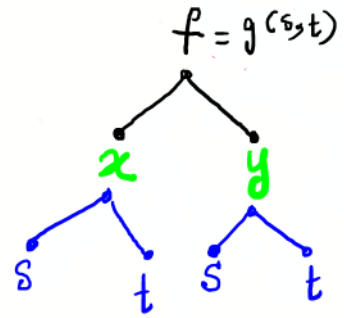
$$= 192$$

Example  $g(s,t) = f(\overset{x}{s^2-t^2}, \overset{y}{t^2-s^2})$ .  $f$  is differentiable.  
show that  $g$  satisfies the equation:

$$t \frac{\partial g}{\partial s} + s \frac{\partial g}{\partial t} = 0$$

Let  $\overset{f(s,t)}{x = s^2 - t^2}$ ,  $\overset{f(s,t)}{y = t^2 - s^2}$

$$\begin{aligned} \therefore g(s,t) &= f(s^2 - t^2, t^2 - s^2) \\ &= f(x, y) \end{aligned}$$



$$\begin{aligned} \frac{\partial g}{\partial s} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s} & \left| \begin{aligned} \frac{\partial g}{\partial t} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t} \\ \frac{\partial g}{\partial s} &= \frac{\partial f}{\partial x} (2s) + \frac{\partial f}{\partial y} (-2t) & \left| \begin{aligned} \frac{\partial g}{\partial t} &= \frac{\partial f}{\partial x} (-2t) + \frac{\partial f}{\partial y} (2t) \end{aligned} \end{aligned} \right. \end{aligned}$$

$$\begin{aligned} \therefore t \frac{\partial g}{\partial s} + s \frac{\partial g}{\partial t} &= \left( 2st \frac{\partial f}{\partial x} - 2st \frac{\partial f}{\partial y} \right) + \left( -2st \frac{\partial f}{\partial x} + 2st \frac{\partial f}{\partial y} \right) \\ &= 0 \end{aligned}$$

OMIT